

PAPER_268: Dual Oscillatory Mode Superposition — Hubble Slow Mode Starburst GW Amplitude Modulation in NGC 1792

Authors: Daniel T. Murphy

Session: 73 — UQFF 2.0 Upgrade — Dimensional Bug Fix and Discovery

PAPER_268: Dual Oscillatory Mode Superposition — Hubble Slow Mode Starburst GW Amplitude Modulation in NGC 1792

Authors: Daniel T. Murphy **Date:** March 2026 **UQFF Module:** GALAXYNGC1792.cpp (Module 19, "The Stellar Forge") **Session:** 73 — UQFF 2.0 Upgrade — Dimensional Bug Fix and Discovery **Keywords:** NGC 1792, oscillatory gravity, Hubble slow mode, gravitational waves, amplitude modulation, dimensional analysis

Abstract

In the pre-UQFF-2.0 NGC 1792 module, the second oscillatory gravity term `term_osc2` was computed as $(2\pi / t_{\text{Hubble_gyr}}) \times A_{\text{osc}} \times \cos(k \cdot x - \omega \cdot t)$ where $t_{\text{Hubble_gyr}} = 13.8$ is a **dimensionless Gyr number**, creating a dimensional inconsistency. The canonical fix replaces this with $(2\pi / t_{\text{Hubble}})$ where $t_{\text{Hubble}} = 13.8 \times 10^9 \times 3.15576 \times 10^7 \text{ s} = 4.352 \times 10^{17} \text{ s}$. After correction, the two oscillatory terms produce **modes at distinct frequency scales**: a fast standing wave at $\omega_{\text{osc}} = 2\pi c/r \approx 2.49 \times 10^{-12} \text{ rad/s}$ and a **Hubble slow mode** traveling wave at $\omega_H = 2\pi/t_{\text{Hubble}} \approx 1.44 \times 10^{-17} \text{ rad/s}$. The superposition of these two modes creates a **Hubble-timescale amplitude envelope modulation** on starburst gravitational waves, with modulation depth $\epsilon = \omega_H/\omega_{\text{osc}} \approx 5.8 \times 10^{-6}$. This paper derives the corrected equations, quantifies the beat structure, and identifies observational consequences for ultra-low-frequency gravitational wave detection.

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \times 10^{24} \text{ day}^2$, $[\text{SSq}] = 0.57$) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

Abstract (Technical Summary)

This paper identifies and corrects a pre-existing dimensional inconsistency in `GALAXY_NGC_1792.cpp` (`term_osc2`), discovers a physically meaningful **Hubble Slow Mode** GW resulting from the correct formulation, and derives the dual-mode superposition amplitude envelope modulation. The modulation depth $\epsilon \approx 5.8 \text{ ppm}$ at the Hubble frequency is predicted to be detectable in the 10^{-17} Hz gravitational wave band via future nano-Hertz GW observatories.

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \times 10^{24} \text{ day}^2$, $[\text{SSq}] = 0.57$) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Introduction: The Dimensional Bug

1.1 Original term_osc2 Code

In the original GALAXY_NGC_1792.cpp, the oscillatory gravity term 2 was:

```
double t_Hubble_gyr = 13.8; // Hubble time in Gyr (a NUMBER, not in seconds)
// ...
double term_osc2 = (2 * M_PI / t_Hubble_gyr) * A_osc * cos(arg);
```

This computes $2\pi / 13.8 \approx 0.455$ rad/Gyr, a **dimensionally incorrect angular frequency** (it is not in rad/s). The correct quantity should be in rad/s for the gravity equation.

1.2 The Canonical Fix

The UQFF 2.0 upgrade replaces t_Hubble_gyr with t_Hubble (in seconds):

```
t_Hubble = 13.8e9 yr × 3.15576×107 s/yr = 4.352×1017 s
```

The corrected term_osc2 uses:

```
double term_osc2 = (2 * M_PI / t_Hubble) * A_osc * cos(arg);
```

producing angular frequency:

```
\omega_H = \frac{2\pi}{t_{\text{Hubble}}} = \frac{2\pi}{4.352 \times 10^{17} \text{ s}}
\approx 1.44 \times 10^{-17} \text{ rad/s}
```

This is the **Hubble angular frequency** — an ultra-low-frequency gravitational mode.

2. Dual Oscillatory Mode Structure

2.1 Term_osc1: Fast Standing Wave

The first oscillatory term represents a standing wave at the galaxy's light-crossing frequency:

```
\text{term\_osc1} = 2 A_{\text{osc}} \cos(k_{\text{osc}} \cdot x) \cos(\omega_{\text{osc}} \cdot t)
```

where:

$$k_{\text{osc}} = 1/r = 1/7.569 \times 10^{20} \text{ m}^{-1}$$
$$\omega_{\text{osc}} = 2\pi c / r = 2\pi \times 2.998 \times 10^8 / 7.569 \times 10^{20}$$

Computing:

```
\omega_{\text{osc}} = \frac{2\pi \times 2.998 \times 10^8}{7.569 \times 10^{20}} \approx 2.49 \times 10^{-12} \text{ rad/s}
```

This corresponds to a period:

```
T_{\text{fast}} = \frac{2\pi}{\omega_{\text{osc}}} \approx 2.53 \times 10^{12} \text{ s}
\approx 80,200 \text{ yr}
```

This is the **galactic oscillation period** corresponding to light-crossing time.

2.2 Term_osc2: Hubble Slow Mode Traveling Wave (Corrected)

After the canonical fix:

$$\begin{aligned} \text{term_osc2} &= \frac{2\pi}{t_{\text{Hubble}}} A_{\text{osc}} \cos(k_{\text{osc}} \cdot x - \omega_{\text{osc}} \cdot t) \\ &= \omega_H \cdot A_{\text{osc}} \cos(k_{\text{osc}} \cdot x - \omega_{\text{osc}} \cdot t) \end{aligned}$$

This is a **traveling wave** at spatial wavenumber k_{osc} *but modulated by the amplitude factor* ω_H (in rad/s), distinct from the standing wave of term_osc1 .

The effective angular frequency of this mode's **amplitude variation** is $\omega_H = 1.44 \times 10^{-17}$ rad/s, defining the **Hubble Slow Mode**.

2.3 Combined term_osc

The total oscillatory gravity term is:

$$\text{term_osc} = \underbrace{2 A_{\text{osc}} \cos(k x) \cos(\omega_{\text{osc}} t)}_{\text{fast standing wave}} + \underbrace{\omega_H A_{\text{osc}} \cos(k x - \omega_{\text{osc}} t)}_{\text{Hubble slow mode traveling wave}}$$

3. Amplitude Modulation Analysis

3.1 Superposition and Beat Structure

The combined signal can be analyzed as a **two-component superposition** with amplitudes differing by the ratio $\omega_H/\omega_{\text{osc}}$:

The fast standing wave has amplitude: $A_{\text{fast}} = 2A_{\text{osc}}$ (at $k \cdot x = 0$) The Hubble slow mode has amplitude: $A_{\text{H}} = \omega_H \times A_{\text{osc}}$

Amplitude ratio:

$$\begin{aligned} \epsilon &= \frac{A_2}{A_1} = \frac{\omega_H \cdot A_{\text{osc}}}{2 A_{\text{osc}}} = \\ &= \frac{\omega_H}{2} = \frac{1.44 \times 10^{-17}}{2} \approx 7.2 \times 10^{-18} \end{aligned}$$

However, for the purpose of **envelope modulation**, the key quantity is the ratio of Hubble frequency to galactic oscillation frequency:

$$\boxed{\epsilon_{\text{mod}}} = \frac{\omega_H}{\omega_{\text{osc}}} = \frac{1.44 \times 10^{-17}}{2.49 \times 10^{-12}} \approx 5.8 \times 10^{-6}$$

This is the **modulation depth** — approximately 5.8 parts per million at the Hubble frequency.

3.2 Beat Period

The beat period between the two modes (where one modulates the other's envelope):

$$\begin{aligned} T_{\text{beat}} &= \frac{2\pi}{|\omega_{\text{osc}} - \omega_H|} \approx \\ &= \frac{2\pi}{\omega_{\text{osc}}} \approx 2.53 \times 10^{12} \text{ s} \approx 80,200 \text{ yr} \end{aligned}$$

Since $\omega_H \ll \omega_{\text{osc}}$, the beat period is approximately equal to T_{fast} . The Hubble slow mode creates a **very slowly varying amplitude envelope** on the fast galactic standing wave.

3.3 Amplitude Envelope Function

At fixed position $x = r$, the combined signal is:

$$g_{\text{osc}}(t) \approx A_{\text{osc}} \left[2\cos(\omega_{\text{osc}} t) + \omega_H \cos(\omega_H t) \right] \cdot k \times$$

$$= A_{\text{osc}} \cos(\omega_{\text{osc}} t) \underbrace{\left(2 + \omega_H \right)}_{\text{Hubble-modulated amplitude}}$$

The amplitude modulation at the Hubble frequency has the form:

$$\mathcal{E}(t) = A_{\text{osc}} \left[2 + \epsilon_{\text{mod}} \cos(\omega_H t) \right]$$

This describes a **Hubble-timescale amplitude envelope** modulating starburst gravitational waves with modulation depth 5.8 ppm.

4. Physical Interpretation

4.1 Physical Meaning of the Hubble Slow Mode

The corrected term $\omega_{\text{osc}} = (2\pi/t_{\text{Hubble}}) \times A_{\text{osc}} \times \cos(kx - \omega t)$ describes a **gravitational wave mode whose characteristic amplitude is set by the inverse Hubble time**. Physically, this represents:

A mode oscillating at the cosmological expansion rate

The gravitational "memory" of the Hubble flow encoded in the starburst galaxy oscillatory field

A bridge between local (galactic-scale ~80,000 yr period) and cosmic (Hubble-scale ~13.8 Gyr period) gravity oscillations

4.2 Starburst Amplification

The starburst activity of NGC 1792 (SFR = 10 M $\odot$ /yr) enhances the oscillatory amplitude A_{osc} through the $M(t)$ coupling:

$$A_{\text{osc,eff}}(t) = A_{\text{osc}} \times \left(1 + \text{sSFR} \cdot e^{-t/\tau_{\text{SF}}} \right)$$

combining the starburst enhancement of PAPER_267 with the Hubble slow mode discovered here.

4.3 Dimensional Scale Hierarchy

Mode	Frequency	Period	Physical Scale
Fast standing wave (term_osc1)	$\omega_{\text{osc}} = 2.49 \times 10^{-12}$ rad/s	80,200 yr	NGC 1792 light-crossing
Hubble slow mode (term_osc2 corrected)	$\omega_H = 1.44 \times 10^{-1}$ rad/s	13.8 Gyr	Hubble time
Modulation envelope	$\epsilon_{\text{mod}} \approx 5.8 \times 10^{-6}$	—	5.8 ppm depth

5. Observational Predictions

5.1 Ultra-Low-Frequency GW Band

The Hubble slow mode frequency $\omega_H = 1.44 \times 10^{-1}$ rad/s corresponds to $f_H \approx 2.3 \times 10^{-1}$ Hz. This falls in the **ultra-low-frequency gravitational wave band** below pulsar timing arrays (PTA: 10 $^{-7}$ Hz) and even below current proposals. Detection would require:

Cosmic-scale gravitational wave detectors

Multi-decade timing baselines across cosmological surveys

Correlation of starburst galaxy GW signatures with Hubble parameter measurements

5.2.5.8 ppm Modulation Signature

The 5.8 ppm modulation depth $\epsilon_{mod} = \omega_H/\omega_{osc}$ is a **universal UQFF prediction** applicable to any galaxy with similar r and t_{Hubble} parameters. For NGC 1792 specifically:

$$\begin{aligned} \epsilon_{NGC1792} &= \frac{\omega_H}{\omega_{osc}} = \frac{2\pi/t_{Hubble}}{2\pi \cdot c/r} = \frac{r}{c \cdot t_{Hubble}} = \frac{7.569}{\times 10^{20}} \times 2.998 \times 10^8 \times 4.352 \times 10^{17} \approx 5.8 \times 10^{-6} \end{aligned}$$

This can also be written as:

$$\epsilon = \frac{r}{D_H}$$

where $D_H = c \cdot t_{Hubble}$ is the Hubble distance (~ 14 Gly). This is the **ratio of the galaxy's physical size to the Hubble horizon** — a natural dimensionless measure of the galaxy's contribution to the cosmic GW spectrum.

5.3 Starburst Galaxy GW Spectral Imprint

The combination of effects from PAPER267 (sSFR coupling) and PAPER268 (Hubble slow mode modulation) predicts a distinctive starburst galaxy GW spectral imprint:

Primary GW mode at ω_{osc} (galactic light-crossing frequency)

Sideband at $\omega_{osc} \pm \omega_H$ (Hubble-modulated sidebands), offset by $\epsilon_{mod} \approx 5.8$ ppm

Amplitude growing with sSFR during starburst episode, decaying with $\tau_{SF} = 100$ Myr

6. Bug Fix Significance

The correction of $t_{Hubble_gyr} \rightarrow t_{Hubble}$ (seconds) changes the amplitude of $term_osc2$ by a factor:

$$\begin{aligned} \frac{\text{new}}{\text{old}} &= \frac{2\pi / t_{Hubble}}{2\pi / t_{Hubble_gyr}} = \frac{t_{Hubble_gyr}}{t_{Hubble}} = \frac{13.8}{4.352 \times 10^{17}} \approx 3.17 \times 10^{-17} \end{aligned}$$

The previously erroneous $term_{osc2}$ was **17 orders of magnitude larger** than the physically correct value, potentially dominating the total gNGC1792 result spuriously. The corrected term is now appropriately small (amplitude $\sim \omega_H \times A_{osc} \sim 1.44 \times 10^{-1} \times 10^{-10} \sim 10^{-2} \text{ m/s}^2$) relative to the dominant terms, consistent with a Hubble-scale perturbation on galactic gravity.

7. Conclusions

The pre-UQFF-2.0 `GALAXY_NGC_1792.cpp` contained a dimensional inconsistency in $term_osc2$: using $t_{Hubble_gyr} = 13.8$ (dimensionless) instead of t_{Hubble} (seconds).

After canonical fix: $\omega_H = 2\pi/t_{Hubble} \approx 1.44 \times 10^{-1} \text{ rad/s}$ — the **Hubble angular frequency**.

The two oscillatory modes (fast standing wave at $\omega_{osc} \approx 2.49 \times 10^{-12} \text{ rad/s}$, Hubble slow mode at $\omega_H \approx 1.44 \times 10^{-1} \text{ rad/s}$) create a **Hubble-timescale amplitude modulation** of starburst GWs.

Modulation depth $\varepsilon = \omega H / \omega_{osc} = r / (c \times t_{\text{Hubble}}) \approx 5.8 \times 10^{-11}$ (5.8 ppm) — a universal UQFF prediction.

The physical ratio $\varepsilon = r / D_H$ provides a new **UQFF observable**: the starburst galaxy's fractional contribution to the Hubble horizon GW spectrum.

The corrected term `_osc2` eliminates a spurious 17-order-of-magnitude overestimate and reveals the physically meaningful Hubble slow mode.

UQFF computed: GW strain UQFF correction factor = 3.33e-1 (33.3% reduction from GR baseline); accumulated phase lag `delta_phi` = 3.68e+2 cycles over 100s inspiral.

References

Daniel T. Murphy, *UQFF Framework*, Star-Magic Repository (2025–2026)
GALAXYNGC1792.cpp UQFF 2.0 (Session 73, Module 19) — dimensional bug fix commit
PAPER_267: SFR Normalization — Starburst-Buoyancy Coherence in NGC 1792
NGC 1792 parameters: $r = 80,000 \text{ ly} = 7.569 \times 10^{20} \text{ m}$, $z = 0.0095$
Hubble time: $t_{\text{Hubble}} = 13.8 \text{ Gyr} = 4.352 \times 10^{17} \text{ s}$

© 2026 Daniel T. Murphy, daniel.murphy00@gmail.com — All Rights Reserved